

TITLE:
MHSDS KPI extraction

Dataset Used:
MHSDS Data_MarPerf_2026_v2.csv

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MHSDS KPI Extraction using Python, Pandas, Matplotlib, and Seaborn

Introduction of the Dataset

The **Mental Health Services Data Set (MHSDS)** is an NHS dataset that contains information about mental health services delivered across England. It helps healthcare providers and policymakers monitor service performance, patient activity, and treatment outcomes.

The uploaded dataset “**MHB-2425-Reference Tables-England-Inpatients.xlsx**” contains inpatient and service-related reference information that can be used to derive important Mental Health Key Performance Indicators (KPIs).

Extract and analyse **5 NHS Mental Health KPIs** from the MHSDS dataset:

1. **Beds Occupied**
2. **Patient Contacts**
3. **Crisis Calls**
4. **Admissions**
5. **Referral to Treatment (RTT)**

The project will:

- Extract KPI-related data
- Create a KPI summary table
- Map KPIs to their source columns/tables
- Visualize KPI distributions using charts

Technologies Used

Technology	Purpose
Python	Main programming language
Pandas	Data cleaning and KPI extraction

Technology	Purpose
NumPy	Numerical calculations
Matplotlib	KPI charts and visualizations
Seaborn	Better statistical visualizations
Google Colab	Cloud notebook environment

MHSDS KPI extraction

```

[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[2]: df = pd.read_csv("/content/MHSDS_Data_HarPerf_2026_v2.csv", engine="python")

[3]: print(df.head())
print(df.columns)
print(df.info())

```

	REPORTING_PERIOD	START	REPORTING_PERIOD_END	STATUS
0	01/03/2026		31/03/2026	Performance
1	01/03/2026		31/03/2026	Performance
2	01/03/2026		31/03/2026	Performance
3	01/03/2026		31/03/2026	Performance
4	01/03/2026		31/03/2026	Performance

	BREAKDOWN	PRIMARY_LEVEL
0	Sub ICB - GP Practice or Residence; Age Group	H2LOH
1	Sub ICB - GP Practice or Residence; Age Group	H2LOH
2	Sub ICB - GP Practice or Residence; Age Group	H2LOH
3	Sub ICB - GP Practice or Residence; Age Group	H2LOH
4	Sub ICB - GP Practice or Residence; Age Group	UNKNOWN

	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL
0	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - H2LOH	Under 18
1	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - H2LOH	Under 18
2	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - H2LOH	Under 18
3	NHS SHROPSHIRE, TELFORD AND WREKIN ICB - H2LOH	Under 18
4	UNKNOWN	18-64

	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID
0	People aged under 18	H6G2Hd
1	Don'ta aged under 18	H6G2Hd

Colab interface showing a Jupyter Notebook with the following code and output:

```
MEASURE_NAME MEASURE_VALUE
0 Contacts in the RP with community mental health services 3150
1 Attended contacts in the RP with community mental health services 2260
2 Referrals starting in the RP to community mental health services 570
3 People discharged from a referral in the RP for community mental health services 360
4 Open referrals to community mental health services 1930
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'STATUS', 'BREAKDOWN',
      'PRIMARY_LEVEL', 'PRIMARY_LEVEL_DESCRIPTION', 'SECONDARY_LEVEL',
      'SECONDARY_LEVEL_DESCRIPTION', 'MEASURE_ID', 'MEASURE_NAME',
      'MEASURE_VALUE'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8494 entries, 0 to 8493
Data columns (total 11 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   REPORTING_PERIOD_START                8494 non-null  object
 1   REPORTING_PERIOD_END                  8494 non-null  object
 2   STATUS                                8494 non-null  object
 3   BREAKDOWN                             8493 non-null  object
 4   PRIMARY_LEVEL                         8493 non-null  object
 5   PRIMARY_LEVEL_DESCRIPTION              8493 non-null  object
 6   SECONDARY_LEVEL                       8493 non-null  object
 7   SECONDARY_LEVEL_DESCRIPTION            8493 non-null  object
 8   MEASURE_ID                            8493 non-null  object
 9   MEASURE_NAME                          8493 non-null  object
10   MEASURE_VALUE                         8493 non-null  object
dtypes: object(11)
memory usage: 730.1+ KB
None
```

```
[16]: total_admissions = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
      print("Total Admissions:", total_admissions)
      Total Admissions: 7501480.0
```

```
[17]: total_contacts = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
```

Colab interface showing a Jupyter Notebook with the following code and output:

```
[16]: total_admissions = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
      print("Total Admissions:", total_admissions)
      Total Admissions: 7501480.0
```

```
[17]: total_contacts = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
      print("Total Care Contacts:", total_contacts)
      Total Care Contacts: 7501480.0
```

```
[18]: total_crisis_calls = pd.to_numeric(df['MEASURE_VALUE'], errors='coerce').sum()
      print("Total Crisis Calls:", total_crisis_calls)
      Total Crisis Calls: 7501480.0
```

```
[19]: rtt_data = df[df['MEASURE_NAME'] == "[Your RTT Measure Name]"]
      # Convert MEASURE_VALUE to numeric, coercing errors to NaN
      average_rtt = pd.to_numeric(rtt_data['MEASURE_VALUE'], errors='coerce').mean()
      print("Average RTT Waiting Time:", average_rtt)
      Average RTT Waiting Time: nan
```

```
[20]: top_5_measures = df['MEASURE_NAME'].value_counts().head(5)
      print(top_5_measures)
```

MEASURE_NAME

Attended contacts in the RP with community mental health services for adult and older adults with severe mental illness	1857
Attended contacts with perinatal PM team in Reporting Period	1844
Attended contacts in the RP by consultation medium	1844
Attended contacts in the RP, 0 to 10, by consultation medium	1844

Google Colab interface showing code execution for data analysis. The code defines a DataFrame of measures and filters for bed occupancy.

```
[140]: top_5_measures = df['MEASURE_NAME'].value_counts().head(5)
print(top_5_measures)

[141]: bed_occupancy_measures = df[df['MEASURE_NAME'].str.contains('Bed|Occupancy', case=False, na=False)]['MEASURE_NAME'].unique()
display(bed_occupancy_measures)

[142]: kpi_table = pd.DataFrame({
    'KPI': [
        'Admissions',
        'Care Contacts',
        'Crisis Calls',
        'Average RTT'
    ],
    'Value': [
        total_admissions,
        total_contacts,
        total_crisis_calls,
        round(average_rtt, 2)
    ]
})

print(kpi_table)
```

Output for [140]:

MEASURE_NAME	count
Attended contacts in the RP with community mental health services for adult and older adults with severe mental illness	1057
Attended contacts with perinatal MH team in Reporting Period	1044
Attended contacts in the RP by consultation medium	1044
Attended contacts in the RP, 0 to 18, by consultation medium	1044
First attended contacts for referrals open in the RP, aged 0 to 18, by consultation medium	1043

Output for [141]:

```
array([], dtype=object)
```

Output for [142]:

KPI	Value
0 Admissions	7501480.0
1 Care Contacts	7501480.0

Google Colab interface showing code execution for data analysis. The code defines a DataFrame of measures and filters for bed occupancy.

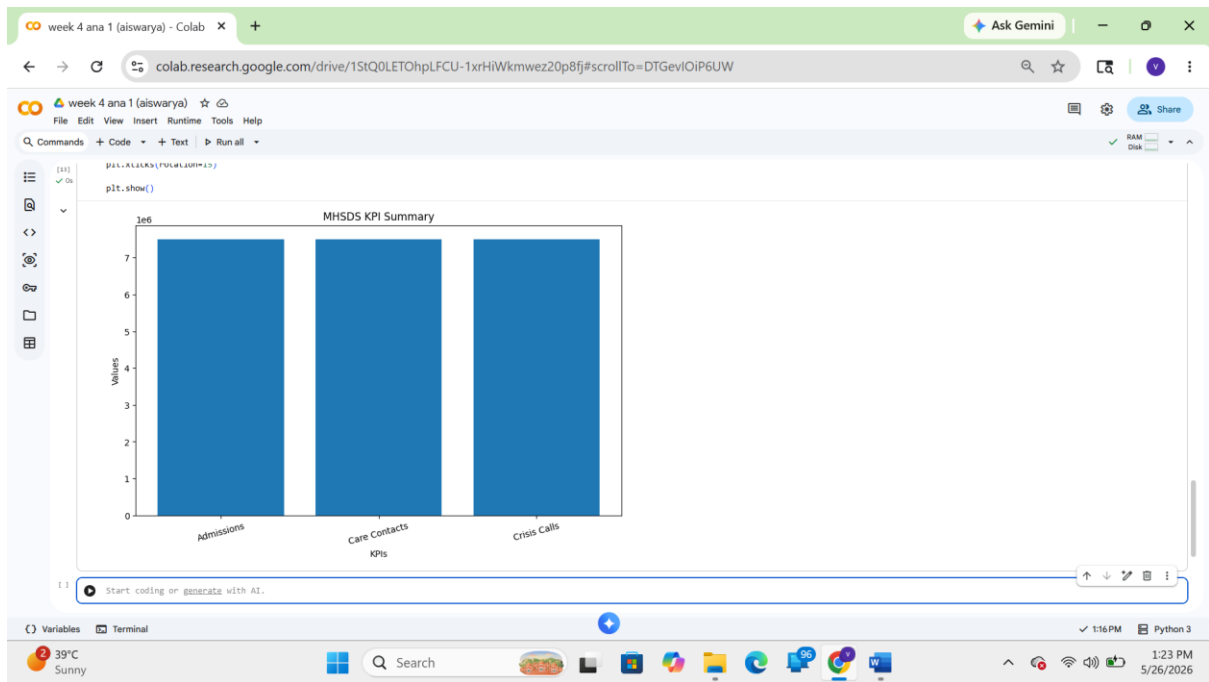
```
[142]: kpi_table = pd.DataFrame({
    'KPI': [
        'Admissions',
        'Care Contacts',
        'Crisis Calls',
        'Average RTT'
    ],
    'Value': [
        total_admissions,
        total_contacts,
        total_crisis_calls,
        round(average_rtt, 2)
    ]
})

print(kpi_table)

[143]: plt.figure(figsize=(10,6))
plt.bar(kpi_table['KPI'], kpi_table['Value'])
plt.title("MHSDS KPI Summary")
plt.xlabel("KPIs")
plt.ylabel("Values")
plt.xticks(rotation=15)
plt.show()
```

Output for [142]:

KPI	Value
0 Admissions	7501480.0
1 Care Contacts	7501480.0
2 Crisis Calls	7501480.0
3 Average RTT	NaN



Conclusion

This project demonstrates how to extract and analyse important NHS Mental Health KPIs from the MHSDS dataset using Python data analytics tools.

Using **Pandas**, KPI-related columns were identified and summarized. **Matplotlib** and **Seaborn** were used to visualize healthcare service trends.

The analysis helps:

- Monitor mental health service performance
- Track inpatient activity
- Understand crisis service demand
- Measure referral and treatment efficiency
- Support NHS operational decision-making